Excess Gaussian energy: standalone supplement

This supplement contains the scalar energy baseline and the excess-energy estimates from *Depth dependence, internal robustness, and Bayesian neural-network learning* (17 October 2026). Depth counts all weighted layers, including the linear readout. The global output margin is 1/2. The calculations concern deterministic parameter energy. The cited simple-target learning result is a separate statistical conclusion of that report, taken at fixed dimension, width and depth before the Gaussian variance tends to zero.

1 Model and scalar transmission baseline

For input dimension $d \geq 1$ and width $N \geq 1$, write

$$h^0 = (1, x), \qquad h^s = \tanh(W^s h^{s-1}) \ (1 \leq s < L), \qquad z_w(x) = v^T h^{L-1}, \qquad x \in \{-1, 1\}^d.$$

There are no later biases. The Gaussian energy is

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

A hidden perturbation η has normalized RMS size $\|\eta\|_2/\sqrt{N}$. A constant-channel network has repeated first-layer bias a, internal entries b_s/N, and readout entries c/N. Its amplitudes and energy obey the scalar optimization below, which is also meaningful for a real first-layer cost coefficient $d > 0$.

Proposition 1.1 (The common depth cost)**.** *For $d > 0$ and integer $L \geq 2$, let $K_*(d, L)$ be the minimum of $da^2 + \sum_{s=2}^{L-1} b_s^2 + c^2$ over nonnegative scalar weights subject to $r_1 = \tanh a$, $r_s = \tanh(b_s r_{s-1})$, and $c r_{L-1} \geq 1/2$. This is the energy of a constant-channel margin-1/2 realization. Put $D = \max\{d, 4\}$. Then*

$$L(d/4)^{1/L} \ \leq \ K_*(d, L) \ \leq \ (L + 12)(D/4)^{1/L}.$$

In particular, for every fixed input dimension, $K_*(d, L) = L + O_d(1)$ as $L \to \infty$. More precisely,

$$\log(d/4) \leq \liminf_{L\to\infty}(K_*(d, L) - L) \leq \limsup_{L\to\infty}(K_*(d, L) - L) \leq 12 + \log(D/4).$$

Proof. Since $\tanh t \leq t$ for $t \geq 0$, scalar feasibility implies $ca \prod_{s=2}^{L-1} b_s \geq 1/2$. The product of the L weighted cost terms $da^2, b_2^2, \ldots, b_{L-1}^2, c^2$ is at least $d/4$. The arithmetic–geometric mean inequality proves the lower bound.

For the upper bound set

$$q = (D/4)^{1/(2L)} \geq 1, \qquad k_s = \min\{s, L - s\}, \qquad r_s = \frac{q^s}{\sqrt{D}\, k_s} \quad (1 \leq s < L).$$

Then $0 < r_s \leq 1/(2qk_s) \leq 1/2$ and $r_{L-1} = 1/(2q)$. Take

$$a = \operatorname{atanh}(r_1), \qquad b_s = \frac{\operatorname{atanh}(r_s)}{r_{s-1}}, \qquad c = q.$$

This realizes the hidden amplitudes r_s exactly and has output 1/2. For $0 < t \leq 1/2$, the power series gives

$$\frac{\operatorname{atanh} t}{t} \leq 1 + \frac{t^2}{3(1 - t^2)} \leq 1 + \frac{4}{9}t^2, \qquad \left(\frac{\operatorname{atanh} t}{t}\right)^2 \leq 1 + t^2.$$

(The last step uses $t^2 \le 1/4$.) Hence

$$da^2 \le dr_1^2 + dr_1^4 \le q^2 + \frac{q^4}{D} \le q^2 + \frac{1}{4}, \qquad c^2 = q^2.$$

Put $R_s = k_{s-1}/k_s$ for $2 \le s < L$. The ratios occur in reciprocal pairs $k/(k+1), (k+1)/k$, with at most one additional ratio equal to one. Each pair contributes at most $4/k^2$ above two, since

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 = \left(\frac{k+1}{k} - \frac{k}{k+1}\right)^2 \le \frac{4}{k^2}.$$

Therefore $\sum_{s=2}^{L-1} R_s^2 \le L - 2 + 4\sum_{k\ge 1} k^{-2} < L - 2 + 7$. Moreover $R_s \le 2$ and $q^{2s+2} \le q^{2L} = D/4$, so

$$\begin{aligned}\sum_{s=2}^{L-1} b_s^2 &\le q^2 \sum_{s=2}^{L-1} R_s^2 + \sum_{s=2}^{L-1} q^2 R_s^2 r_s^2 \\ &\le q^2(L-2+7) + \sum_{s=2}^{L-1} \frac{1}{k_s^2} < q^2(L-2+7) + 4.\end{aligned}$$

Adding the endpoint costs gives $K_*(d, L) \le q^2(L+7) + 17/4 \le q^2(L+12)$. The argument also applies at $L = 2$, when the internal sums are empty. Finally, expand $Le^{\log(d/4)/L}$ and $(L+12)e^{\log(D/4)/L}$ for fixed d to obtain the asymptotic statements. □

2 What excess Gaussian energy does and does not control

Write a width-N, depth-L tanh network as $h^1 = \tanh(W^1\bar{x})$, $h^s = \tanh(W^s h^{s-1})$ for $2 \le s < L$, and $z = v^T h^{L-1}$, where $\bar{x} = (1, x)$ and $x \in \{-1, 1\}^d$. Its weighted Gaussian energy is

$$\kappa_N = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2.$$

The following facts concern this actual energy. They do not establish a sample-complexity lower bound for the unrestricted Gaussian posterior.

Proposition 2.1 (Excess energy controls total deviation from rank one)**.** *Suppose $|z(x)| \ge 1/2$ on the whole Boolean cube. Let R_s be a best rank-one Frobenius approximation to W^s, and let $\rho_s = \|W^s\|_F^2/\|W^s\|_{\mathrm{op}}^2$ be its stable rank. Then*

$$\sum_{s=2}^{L-1} \|W^s - R_s\|_F^2 \le \kappa_N - L(d/4)^{1/L}, \tag{1}$$

$$\sum_{s=2}^{L-1} \log \rho_s \le L \log(\kappa_N/L) - \log(d/4). \tag{2}$$

In particular, if $\kappa_N \le L + B$, both right-hand sides are at most $B - \log(d/4)$.

Proof. Set $a = \|W^1\|_F/\sqrt{N}$, $c = \sqrt{N}\|v\|_2$, and $p_s = \|W^s\|_{\mathrm{op}}$. Cube averaging supplies an input with $\|W^1\bar{x}\|_2/\sqrt{N} \le a$. Tanh contraction gives $1/2 \le ac\prod_s p_s$. Thus the product of the L numbers $da^2, p_2^2, \ldots, p_{L-1}^2, c^2$ is at least $d/4$, and their sum is at least $L(d/4)^{1/L}$. Subtract this sum from κ_N and use $\|W^s - R_s\|_F^2 = \|W^s\|_F^2 - p_s^2$ to obtain (1). Alternatively, the product of the Frobenius energy terms is at least $(d/4)\prod_s \rho_s$ and at most $(\kappa_N/L)^L$, which proves the logarithmic bound. Finally use $e^u \ge 1 + u$ and $\log(1+u) \le u$. □

Thus bounded excess permits only a bounded total squared residual after replacing internal matrices by rank-one matrices. This does not justify replacing those matrices in the computation: the suffix can amplify the replacement errors. The next example shows that even a constant excess allowance does not imply inverse-polynomial hidden-state robustness.

Proposition 2.2 (Bounded excess permits a fragile scalar transmission)**.** *For $d = 4$, every even $L \geq 4$, and every $N \geq 1$, there is a network with $z(x) = 1$ for all inputs and*

$$\kappa_N \leq L + 76 \leq K_*(4, L) + 76,$$

whose output can be made zero by a perturbation at one hidden layer of normalized RMS at most $L^{-1}e^{-\sqrt{L}/4}$. Here $K_*(4, L)$ is the scalar minimum for margin-$1/2$ transmission defined above. It also equals the globally minimum margin-$1/2$ energy for this model, by the simple-target result in the research report.

Proof. Put $n = L/2$, $h = L^{-1/2}$, $k_s = \min(s, L - s)$, and

$$r_s = \frac{e^{-h(k_s-1)}}{2k_s}, \qquad 1 \leq s < L.$$

Give each first-layer row the sole constant-input weight $\operatorname{atanh}(1/2)$. Let every entry of W^s be b_s/N, where $b_s = \operatorname{atanh}(r_s)/r_{s-1}$, and let $v_i = 2/N$. The hidden vector is exactly $r_s\mathbf{1}$, so the output is $2r_{L-1} = 1$ and the energy is

$$4\operatorname{atanh}(1/2)^2 + \sum_{s=2}^{L-1} b_s^2 + 4.$$

For completeness, the following loose constants are uniform in depth. The ratios r_s/r_{s-1} occur in reciprocal pairs t_k, t_k^{-1}, where $t_k = (1 + 1/k)e^h$ and $1 \leq k < n$. Since $h \leq 1/2$,

$$t_k - t_k^{-1} \leq e^h(2/k + 2h), \qquad t_k^2 + t_k^{-2} - 2 \leq 8e^{2h}(k^{-2} + h^2).$$

Using $e^{2h} < 3$, $\sum_{k\geq 1} k^{-2} < 2$, and $(n-1)h^2 \leq 1/2$ gives $\sum_s (r_s/r_{s-1})^2 < L - 2 + 60$. Also $\max_s (r_s/r_{s-1})^2 < 12$ and $\sum_s r_s^2 < 1$. For $0 < t \leq 1/2$ the power series gives $(\operatorname{atanh}(t)/t)^2 \leq 1 + t^2$. Consequently

$$\sum_s b_s^2 < L - 2 + 72, \qquad 4\operatorname{atanh}(1/2)^2 \leq 5/4, \qquad \kappa_N < L + 76.$$

The baseline lower bound gives $K_*(4, L) \geq L$. Finally, adding $-r_n\mathbf{1}$ after the middle activation zeros that state and hence every later state. Its normalized RMS is

$$r_n = L^{-1}\exp\left(-\frac{L/2 - 1}{\sqrt{L}}\right) \leq L^{-1}e^{-\sqrt{L}/4}.$$

□

This example represents a statistically easy target: the report's simple-target theorem gives depth-independent strong-prior learning for constants, at fixed width and depth before taking the variance limit. It therefore separates fragility of a cheap parameter realization from difficulty of learning its Boolean function. It does not show that the fragile realization is a minimum-energy representative.

Proposition 2.3 (Additional depth has asymptotically unit energy cost)**.** *Suppose a width-N, depth-L_0 network realizes f with margin at least $1/2$. Write $c^2 = N\|v\|_2^2 > 0$ and $E_{\mathrm{pre}} = \kappa_N - c^2$. For every integer $t \geq 1$, there is a width-N, depth-$(L_0 + t)$ realization of the same function, with margin at least $1/2$ and energy at most*

$$E_{\mathrm{pre}} + (t + 13)\max\{c^2, 1\}^{1/(t+1)}.$$

Write $\kappa_{N,L}(f)$ for the infimum energy of depth-L, width-N representations with signed output strictly greater than $1/2$ on the cube. For each fixed representable f, d, N, this infimum satisfies $\kappa_{N,L}(f) = L + O_{f,d,N}(1)$ as $L \to \infty$.

Proof. Extend the scalar minimum $K_*(D, m)$ to any real $D > 0$: minimize $Da^2 + \sum_{j=2}^{m-1} b_j^2 + c'^2$ subject to $c' r_{m-1} \geq 1/2$, where $r_1 = \tanh a$ and $r_j = \tanh(b_j r_{j-1})$. The scalar baseline upper-bound argument works unchanged for real $D > 0$:

$$K_*(D, m) \leq (m + 12)(\max\{D, 4\}/4)^{1/m}.$$

Indeed, setting $q = (\max\{D, 4\}/4)^{1/(2m)}$ and $r_j = q^j/[\sqrt{\max\{D, 4\}}\min(j, m - j)]$ gives this bound by $a = \operatorname{atanh} r_1$, $b_j = \operatorname{atanh}(r_j)/r_{j-1}$, $c' = q$, and the summable reciprocal-ratio estimates used earlier.

Take the scalar tuple for $D = 4c^2$ and $m = t+1$. Replace the old readout by a hidden layer whose every row is $2av^T$, followed by the remaining scalar layers and readout. Its first new preactivation is $2az(x)$, whose absolute value is at least a and whose sign is $f(x)$. Monotonicity and oddness give the same sign with final margin at least $1/2$. The first new block has energy $4a^2c^2$; the remaining new blocks have exactly the remaining scalar costs. The total is therefore $E_{\mathrm{pre}} + K_*(4c^2, t + 1)$, proving the displayed upper bound. An arbitrarily small readout increase supplies strict margins. Expanding this bound as $t \to \infty$, and combining it with $L(d/4)^{1/L} \leq \kappa_{N,L}(f)$, proves the final assertion. □

These energy estimates alone do not settle unrestricted Gaussian sample complexity. They rule out exponential growth of minimum target energy under depth padding, while showing that bounded excess does not make every cheap realization robust. A target-relative argument must also control competing functions and their prior masses; subtracting a common transmission cost alone does not do so.